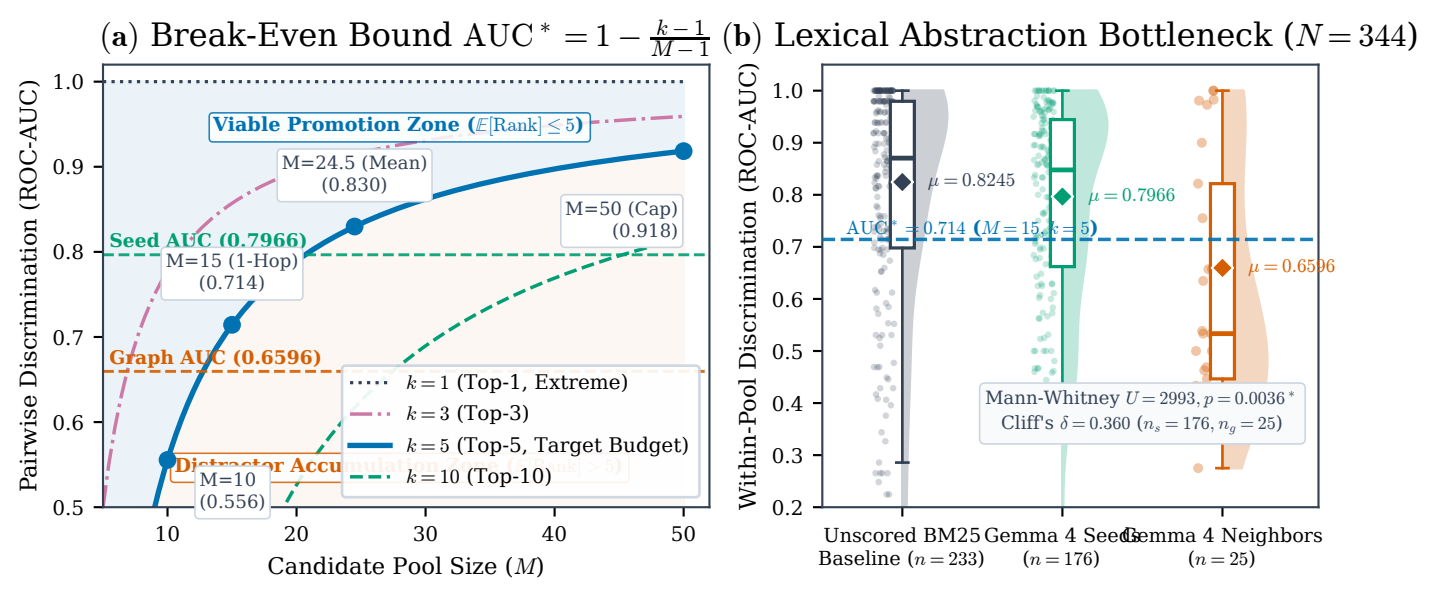

(a) Break-Even Bound $\mathrm{AUC}^* = 1 - \frac{k-1}{M-1}$
Viable Promotion Zone ($\mathbb{E}[\mathrm{Rank}] \leq 5$)
M=24.5 (Mean)
(0.830)
M=50 (Cap)
(0.918)
Seed AUC (0.7966)
M=15 (1-Hop)
(0.714)
Graph AUC (0.6596)
M=10
(0.556)
$k = 1$ (Top-1, Extreme)
$k = 3$ (Top-3)
$k = 5$ (Top-5, Target Budget)
$k = 10$ (Top-10)
Pairwise Discrimination (ROC-AUC)
Candidate Pool Size (M)
(b) Lexical Abstraction Bottleneck ($N = 344$)
$\mu = 0.8245$
$\mu = 0.7966$
$\mu = 0.6596$
$\mathrm{AUC}^* = 0.714$ ($M = 15, k = 5$)
Mann-Whitney $U = 2993, p = 0.0036^*$
Cliff's $\delta = 0.360$ ($n_s = 176, n_g = 25$)
Within-Pool Discrimination (ROC-AUC)
Unscored BM25 Baseline ($n = 233$)
Gemma 4 Seeds ($n = 176$)
Gemma 4 Neighbors ($n = 25$)